

ADELARD DCUNHA

AI/ML Engineer | Generative AI, Agentic Systems, RAG

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SUMMARY

AI/ML engineer, building generative and agentic AI that runs in production. At EduvanceAI that means multi-agent systems letting consultants query their own datasets in plain language, RAG pipelines over documents and video, and the chat and voice layers sitting on top of them. A good share of the work is the part nobody demos: sandboxing code the agents write, keeping every answer traceable back to a source page, and watching models for drift after they ship. Also built a real-time card fraud detection service, which is my closest brush with BFSI.

SKILLS

Agentic AI: Multi-agent systems, agent planning and intent decomposition, semantic memory, tool use and tool generation, sandboxed execution, guardrails, LangGraph

Generative AI and LLMs: RAG, prompt engineering and evaluation, fine-tuning, document understanding, conversational AI, summary generation, quantization, inference and cost optimization

Retrieval and storage: Pinecone, ChromaDB, Neo4j knowledge graphs, Redis, PostgreSQL, multimodal retrieval, recall@k evaluation

ML, deep learning and NLP: Classification, regression, clustering, anomaly detection, time series forecasting, speech-to-text, feature engineering, model evaluation

Languages and frameworks: Python, SQL, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, Pandas, NumPy, SHAP, Node.js

Data and infrastructure: PySpark Structured Streaming, Apache Kafka, Docker, Kubernetes, AWS (EC2, SageMaker), REST APIs and microservices

MLOps: MLflow, drift detection (PSI and KS), automated retraining, champion-challenger promotion, shadow and canary rollout, model monitoring

EXPERIENCE

Software Development Engineer, EduvanceAI

Feb 2025 to present

- Built a multi-agent analytics platform that lets consultants ask questions of their own datasets in plain English. A custom datadictionary layer gives the agents schema context, and a memory component pulls in relevant history as the conversation runs. It handles feature engineering, EDA and analysis, and accepts voice input. Now used by 100+ people across 4 enterprise clients.
- Wrote a tool generator that writes Python tools on the fly and runs them in a sandbox. Breaking each query into a list of intents took output consistency on repeated questions from 62% to 100%. The sandbox plus static checks mean agent-written code cannot execute anything arbitrary. Also built the Next.js chat frontend.
- Built the RAG pipelines over 1,000+ hours of video and 1,200 documents, with Neo4j holding the knowledge graph and Pinecone the vectors. Video answers come back with timestamps and document answers with page numbers, so a field technician can check the source rather than take the model's word for it.
- Researched and wrote an audio-processing library for the voice assistant: voice activity detection, silence trimming, noise cancellation, clipping repair, click and pop removal, GPU acceleration. Throughput went up 40% and cost per minute fell around 60%. I also produced the WER-versus-latency numbers we used to settle on the streaming setup.
- Cut embedding spend from \$512 a month to roughly \$80. Benchmarked the OpenAI embedding API against a self-hosted quantized multilingual model on an EC2 G4 instance using WER and recall@k, then moved Hindi, English and Hinglish retrieval across to it.
- Shipped Sales Copilot: forecasting, dealer clustering and TSP route optimization that produces monthly journey plans for 70 outlets across South and East India. Planned travel distance dropped 70% and visit coverage rose 50%. Added collaborative filtering for next-best-product, and a Next Best Action pulled from call audio.
- Built a real-time error-logging and model-monitoring microservice. Downtime fell 99%, from near-constant to about once a week, and we now get to root cause roughly twice as fast.

- Earlier on: role-based access control (super admin, admin, user), an SMTP notification service, and the Eduvance.ai marketing site for a soft-skills LMS with voice and chat built in.

PROJECTS

Real-time fraud and anomaly detection for card transactions (BFSI)

- A flagged card transaction has to be stopped before authorization finishes, so every event is scored inline. p99 stays under 40ms against a Redis online feature store.
- Two models rather than one: gradient boosting for fraud patterns already seen, and an autoencoder reconstruction error for the ones that are not. Positive class is 0.2%.
- Thresholds come from expected-cost curves weighing chargeback losses against customers lost to false declines. Optimizing F1 would have picked the wrong operating point.
- Features are built in PySpark Structured Streaming with point-in-time correctness. An earlier offline version had label leakage worth 11 AUC points.
- PSI and KS drift monitors trigger retraining, MLflow gates promotion on precision@k, and new models go out shadow-mode then canary. Every alert carries a SHAP explanation so an analyst can audit the decision.

Autonomous voice intake and triage agent for service providers

- An agent that answers the phone for appliance repair businesses and runs the whole intake: takes the complaint, works out what is wrong, either resolves it or escalates, and books the visit. Calls that used to ring out after hours now get a response.
- Speech-to-text and intent classification run on the live transcript, separating problems the caller can fix themselves from ones that need a technician on site.
- For the minor cases an LLM layer suggests likely causes, a DIY fix, or a replacement part. Anything complicated gets escalated to a scheduled visit.
- Appointments are booked and confirmed straight from the transcript and synced to the dispatch calendar, with nobody in the loop. Runs on a telephony and VoIP backend that handles concurrent inbound calls.

Hierarchical demand forecasting for multi-store retail inventory

- A retailer that replenishes per store but budgets per region needs forecasts that agree with each other. MinT reconciliation makes store-level predictions sum exactly to regional and national totals, which got rid of the manual overrides planners were using to paper over the mismatch.
- Several thousand intermittent series. Rolling-origin backtesting against seasonal-naive and gradient-boosted baselines before trusting the Keras model over either.
- Conformal prediction supplies calibrated intervals, so safety stock comes from measured uncertainty instead of a fixed buffer.
- Weekly SageMaker retraining with champion-challenger promotion, gated on a statistically significant pinball-loss improvement. Drift detection falls back to the statistical baseline when the neural forecasts start degrading.

EDUCATION

- **BE, Artificial Intelligence and Data Science**, Fr. Conceicao Rodrigues College of Engineering, Mumbai. 2021 to 2024.
- **Diploma, Electronics and Telecommunication**, St. Xavier's Technical Institute, Mumbai. 2018 to 2021.

CERTIFICATIONS

- **Mathematics for Machine Learning and Data Science**, DeepLearning.AI on Coursera
- **Machine Learning Specialization**, DeepLearning.AI and Stanford University on Coursera

WRITING

- Your ASR Model Isn't the Problem, Your Signal Path Is. Medium: medium.com/@adelarddcunha07